UDCT v7.07

Post-v7.06: structural tests of reciprocal projection, three-dimensional actions, and Solar-System limits

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Abstract

Starting from the v7.06 result that the Hessian-dependent action derivative dominates three dwarf-galaxy virial tests, this note asks whether an action-consistent reciprocal field can avoid that stress while retaining a distinct, testable gravitational response. Five structural controls are reported. A first-gradient QUMOND-form action removes the v7.06 double-divergence term and restores positive conditional virial work, but its stipulated stellar dispersions remain too low. A stable, linearly coupled massive field produces a Yukawa correction that weakens with distance. An exact three-dimensional-to-two-dimensional planar projection gives flat outer speeds but the wrong baryonic mass exponent. A three-dimensional nonlinear scalar restores the deep-field mass exponent, yet its polynomial high-field completion predicts an enormous extra Mercury perihelion precession. A saturated completion suppresses the isolated-Sun radial residual, correcting that over-broad negative inference, but uses the pre-existing standard MOND interpolation and has not passed an external-field, lensing, or independent-projection test.

Status and interpretation

All action choices below were formulated **after** the historical 0/22 preregistered O8d velocity-proxy result. The new numbers are synthetic, analytic, or conditional calculations, not held-out fits or empirical validation. "Bidirectional" means reciprocal coupling derived by varying one action; it does not by itself establish a physically unique UDCT projection. The v7.06 field and Jeans results, the v5.61 archived gate, and the original 1-D feedback caveats remain unchanged. [1-3]

The structural question

The old action depended explicitly on a Newtonian-field Hessian H. Its variation contains a tensor response P_{ij} followed by $\partial_i\partial_j P_{ij}$. In the v7.06 configurations that contribution was numerically much larger than the wanted force. This establishes an oversized response in those models, **not** a proof of an infinite mathematical divergence. The following actions use first spatial derivatives of independent fields, and each is varied consistently before testing its implications.

1. First-gradient control: removing the Hessian stress

$L = \rho(v^2/2 - \Phi) - [2\nabla\Phi\cdot\nabla U - a_0^2 Q(z)]/(8\pi G)$, $z = |\nabla U|^2/a_0^2$; $Q_z(z) = (1 + 1/z)^{1/4}$.

$\nabla^2 U = 4\pi G\rho$; $\nabla^2\Phi = \nabla\cdot[\nu(|\nabla U|/a_0)\nabla U]$; $\nu(x) = (1 + x^{-2})^{1/4}$.

This explicit two-potential action has the field-equation structure of QUMOND [4], with a chosen interpolation that approaches $\nu \sim x^{-1/2}$ for x much smaller than 1 and $\nu - 1 \sim 1/(4x^2)$ for x much larger than 1. Its field equation has no action-Hessian P tensor. Near a zero of ∇U the constitutive derivative still requires careful weak-solution and numerical treatment. This is a structural control, not a new UDCT law.

The same visible Galactic disk and bulge E,H, frozen stellar M/L=2, catalog half-light radii and Plummer/tapered-cusp choices used in v7.05-v7.06 were held fixed. Only the constitutive field equation was changed. The virial number is density-weighted work divided by that profile's Newtonian work. The LOS number follows a conditional spherical, isotropic, angular-mean Jeans calculation in a projected R_e cylinder, with zero outer pressure; it is not a 3-D distribution function.

Object	Virial: Plummer / cusp	LOS at Re: Plummer / cusp	Archived velocity
Leo IV	13.935 / 13.861	1.122 / 1.188 km/s	3.3 km/s
Pegasus III	19.321 / 19.192	0.976 / 1.032	5.4
Hydrus I	3.626 / 3.423	0.595 / 0.627	2.7

Virial radial/angular grids 180×16×24 and 360×24×36 agree to roughly 4×10^{-6} absolute in the largest tapered-profile difference; 450×16×24 and 900×24×36 Jeans grids agree to about 0.0004 km/s or better. Positive work does not imply a fit. Observed velocity selections and aperture definitions were not recreated, and the nonaxisymmetric field was angular-averaged. The archived O8d result is untouched.

Decision

Removing Hessian dependence can remove this particular derivative-stress obstruction, but a smooth action plus positive virial work is insufficient: absolute stellar speed, external-field boundary, and independent prediction still fail to close. The present construction is a MOND-family control, not demonstrated bidirectional novelty.

2. Reciprocal linear field: stability does not ensure the right radial trend

$$L = \rho(v^2/2 - \Phi) + \{[(\partial_t B)^2/c_B^2] - |\nabla\Phi|^2 - |\nabla B|^2 + 2\alpha\nabla\Phi\cdot\nabla B - m^2B^2\}/(8\pi G).$$

$$\nabla^2\Phi - \alpha\nabla^2 B = 4\pi G\rho;\ \nabla^2 B - \alpha\nabla^2\Phi - m^2 B - (\partial_t^2 B)/c_B^2 = 0.$$

This reversible toy lets matter source Φ, Φ source B, and B feed back into Φ and matter. In its linear vacuum mode, $\omega^2 = c_B^2[(1 - \alpha^2)k^2 + m^2]$; the mode has positive frequency squared for $|\alpha| < 1$ and $m^2 >= 0$. This necessary linear stability check is not a complete relativistic or energy-positivity proof. For a static point mass the Fourier field can be resolved exactly:

$$\Phi(r) = -GM[1 + \beta\exp(-r/\lambda)]/r;\ \beta = \alpha^2/(1 - \alpha^2),\ \lambda = \sqrt{(1 - \alpha^2)}/m.$$

$$g/g_N = 1 + \beta(1 + r/\lambda)\exp(-r/\lambda).$$

r / λ	Boost for illustrative β = 1	Trend
0.01	1.99995	Strongest close to source
1	1.73576	Already decreasing
10	1.00050	Approaches Newtonian

The derivative of (1+t)exp(-t) is -t exp(-t). Thus the added boost decreases with radius for every positive stable β. Taking m=0 leaves only a constant rescaling of Newtonian G. β=1 merely illustrates the exact theorem; it was not fitted. Such Yukawa-type forces are familiar in massive-field gravity models. [5] Reciprocity alone is not the distinctive large-radius response sought by UDCT.

Comparison to archived 1-D B feedback

The v5.32 toy prescribed $B_t = A\partial_x^2\rho - B - B^3/2$, while particles felt $-\partial_x\Phi$ through a separate B-dependent interpolation. It showed an on/off trajectory difference but no joint field-particle action. The density-Laplacian source could arise as a dissipative gradient flow from an interaction $-A\int B\partial_x^2\rho\, dx$. Varying that same interaction with respect to particle density also creates a reciprocal B-curvature force, schematically $+A\partial_x^3 B$ in the particle acceleration. That reciprocal force was not the prescribed v5.32 particle equation. This natural-action comparison does **not** prove no more elaborate open-system action exists; it identifies what a new derivation must explain. [2,3]

3. Literal 3-D to 2-D projection: flat speed, incorrect mass scaling

$\Sigma(X) = P\rho = \int \rho(X,z)dz$; $S_B = \int dt[-\int d^2X\, |\nabla_\perp B|^2/(2\kappa) - \int d^3x\, \rho(x)B(X)]$.

$\nabla_\perp{}^2 B = \kappa\Sigma$; $a_{B,\perp} = -\nabla_\perp B$; $a_{B,z} = 0$.

The first equation follows by varying B, and the force by varying matter in the same interaction. In operator language the return map is $P^\dagger B = B(X)$: a genuine reciprocal forward/adjoint pair. κ is a new dimensionful coupling that this calculation does not derive. A fixed plane is chosen by hand, so full 3-D rotational symmetry and a natural direction for a spheroidal dwarf are also absent.

Outside compact projected mass M: $B = [\kappa M/(2\pi)]\ln R + \text{constant}$; $g_B = \kappa M/(2\pi R)$; $v_{flat}{}^2 = \kappa M/(2\pi)$; $v_{flat}{}^4 \propto M^2$.

The logarithmic 2-D Green function makes the outer speed flat, but the observed baryonic Tully-Fisher scaling is approximately $v_{flat}{}^4 \propto M$, not M^2. [6] For a transparent example only, calibrating one hypothetical $M_0 = 10^{10}$ solar masses to the frozen $\xi_0 = 1.082401\times10^{-10}$ m s^{-2} fixes $\kappa = 2\pi\sqrt{(G\xi_0/M_0)}$. The mismatch away from M_0 cannot be removed with that universal κ.

M / M0	Planar speed (km/s)	Desired sqrt(M) law (km/s)	v^4 / (GM ξ0)
0.01	10.948	34.620	0.01
1	109.478	109.478	1
10	346.201	194.683	10

As a finite-radius check, an exponential surface density encloses the fractions 0.5940, 0.9596, and 0.9995 of M at 2, 5 and 10 disk scale lengths. Thus the planar contribution can become nearly flat by 5-10 lengths; its mass exponent remains wrong. This example calibration is diagnostic, not an observed fit.

Simple power-law field generalization

For an isolated two-dimensional p-Laplacian, $\nabla_\perp \cdot (|\nabla B|^{p-2}\nabla B) = \kappa\Sigma$, Gauss flux implies $v^2 \propto M^{1/(p-1)} R^{(p-2)/(p-1)}$. Flat speed demands p=2, which returns $v^2 \propto M$. p=3 gives the desired √M mass factor in v^2 but predicts $v^2 \propto \sqrt{R}$, a rising curve. This excludes this narrow local, single-power, fixed-plane class. It is not a theorem against every nonlocal or dynamical projection.

4. Three-dimensional nonlinear B: a successful deep limit, a failed high limit

$S_B = \int dt\, d^3x[-a_*^2 F(y^2)/(8\pi G) - \rho B]$, $y = |\nabla B|/a_*$; $F(y^2) = (2/3)y^3 + (2/5)y^5$.

$\nabla\cdot[(y+y^3)\nabla B] = 4\pi G\rho$; $a_{matter} = -\nabla(\Phi_N + B)$.

For the point-source and isolated-Sun checks here, $a_*=\xi_0$ and G is the measured Newtonian value: at large source acceleration this polynomial scalar becomes a small fractional correction rather than a Newtonian rescaling. With $g_N=GM/r^2$, the exact spherical relation is $y^2+y^4=g_N/a_*$. Deep outside, $|\nabla B|\approx\sqrt{(GMa_*)}/r$, yielding $v^4\approx GMa_*$. Tangential and radial static linearization coefficients are $y+y^3$ and $2y+4y^3$, respectively, both positive for $y>0$. At $y=0$ the equation is degenerate, not uniformly elliptic.

At high source acceleration the same law has $|\nabla B|\approx a_*(g_N/a_*)^{1/4}$. It remains a distance-dependent extra force in the planetary region. A Kepler-orbit first-order Gauss integral was evaluated at the standard Mercury, Earth and Saturn semimajor axes and eccentricities:

$\Delta\varpi_B = [1/(eGM_{Sun})]\int_0^{2\pi} b(r(f))r(f)^2\cos(f)df$; $r(f)=a(1-e^2)/(1+e\cos f)$.

Object	Extra b at semimajor axis	Conditional precession
Mercury	1.497×10^-8 m/s²	-149.972 arcsec/century
Earth	9.312×10^-9 m/s²	-152.609 arcsec/century
Saturn	3.007×10^-9 m/s²	-153.426 arcsec/century

For Mercury, 257, 1,025, 4,097 and 16,385 anomaly-grid samples agree to the shown precession digits. This perturbative extra secular effect alone is far larger, and of opposite sign, than the familiar +42.98 arcsec/century GR contribution; a full ephemeris refit was not performed. [7] The polynomial completion is therefore not viable as written. The result does **not** exclude all first-gradient B actions; the next construction tests a different high-field asymptote without selecting a fit to Mercury.

5. A saturated first-gradient completion: a necessary correction

$F(y^2) = y\sqrt{(1+y^2)} - \operatorname{asinh}(y)$; $\mu_B(y) = dF/d(y^2) = y/\sqrt{(1+y^2)}$.

$\nabla\cdot[\mu_B(y)\nabla B] = 4\pi G_{bare}\rho$; $g_{total} = g_{bare} + |\nabla B|$.

The standard MOND interpolation $\mu=y/\sqrt{(1+y^2)}$ was already in the literature [8]. Here it is a control in an additive Newton-plus-B action, not a newly derived UDCT function. For a point source its spherical relation is $y^2/\sqrt{(1+y^2)}=g_{bare}/a_*$. In the strong field, $|\nabla B|\approx g_{bare}+a_*^2/(2g_{bare})$. Therefore the leading local gravitational constant is $2G_{bare}$. Set $G_{bare}=G_{lab}/2$ to match its measured value. To retain the frozen deep coefficient $v^4=G_{lab}M\xi_0$, set $a_*=2\xi_0$. These factors follow algebraically from the selected action and both normalizations; they do not imply that the action itself was derived from UDCT physics.

High-field non-Newtonian residual: $\delta g = |\nabla B| - g_{bare} \approx a_*^2/(2g_{bare}) = 4\xi_0^2/g_{lab}$.

Planet	Isolated-Sun δg at a	Added perihelion shift
Mercury	1.184×10^-18 m/s²	-3.185×10^-8 arcsec/century
Earth	7.903×10^-18 m/s²	-3.454×10^-7 arcsec/century
Mars	1.835×10^-17 m/s²	-9.877×10^-7 arcsec/century

The exact implicit spherical solution, numerically stable after subtraction of the absorbed g_{bare} term, was inserted into the same first-order Gauss integral. These are **isolated-Sun radial residuals**, not a full Solar-System pass. Static ellipticity coefficients are $\mu_B>0$ and $\mu_B+y\mu_B'>0$ for $y>0$, but the zero-field limit remains degenerate. Solar Galactic external-field quadrupoles, multiple bodies, and a relativistic light-coupling metric were not computed. Updated Cassini radio tracking particularly constrains MOND-like external-field quadrupoles. [9]

Corrected inference

The very large Mercury precession in Section 4 came from that polynomial choice of high-field response. It is not a no-go theorem for every single-B action. Saturating μ removes that isolated-radial failure, but simultaneously reveals the lack of UDCT-specific content: the successful control is a known MOND-family interpolation. It cannot be counted as a distinct bidirectional result.

6. What remains specifically testable as UDCT projection?

A nontrivial projector must be physically defined rather than appended to an already successful MOND law. A candidate interaction has the general form $S_{int} = -\int B(P\rho)$, with a definite map P from physical matter to the B source. Variation with respect to matter then fixes the adjoint backreaction $P^{\dagger}B$. This is a reciprocity constraint. If P is merely the identity, the model reduces to an ordinary auxiliary scalar. A literal fixed-plane P fails the mass-scaling and vertical-force gates in Section 3. Arbitrarily fitting a length, transition switch, or projection orientation after seeing the data would erase the intended test.

For any normalized, finite-range, translation-invariant smoothing P that conserves total source mass, the leading far-field monopole is still M: $\int P\rho\, d^3x = \int \rho\, d^3x$. A deep 3-D B law with the same asymptote therefore retains the same leading $v^4 \propto M$ prediction as its P=identity control. Any distinctive projection signature must be derived in finite-radius, angular, time-dependent, environmental, or lensing observables, and may demand an independently justified new scale. This statement is restricted to this normalized linear smoothing class.

Tested structure	Positive structural result	Blocking issue
QUMOND-form first-gradient control	No P_ij double divergence; positive virial	Known framework; conditional velocities low
Massive reciprocal B	Stable linear mode for |alpha|<1	Yukawa boost weakens outward
Exact planar projector	Reciprocal action; flat outer speed	v^4 scales as M^2; no vertical B force
Polynomial 3-D B	Deep v^4 scales as M; elliptic for y>0	About -150 arcsec/cy Mercury addition
Saturated 3-D B	Tiny isolated-Sun radial residual	Standard MOND function; Q2/lensing open

Next gates, in order

First: specify a geometry-based, observer-independent P and its adjoint; derive both maps and a translation/rotation or full field-momentum conservation statement from one action. **Second:** solve a finite 3-D matter distribution and external Galactic field, with zero-coupling and P=identity controls. Require a distinctive, frozen statistic above numerical and baryonic-map uncertainties. **Third:** test the Solar-System external-field quadrupole and planetary multi-body dynamics, then link the same action to a photon metric and lensing. **Fourth:** only after those gates, perform matched-aperture galaxy and dwarf velocity comparisons against AQUAL/QUMOND and standard gravity. The historical v5.61 O8d 0/22 result remains a separate frozen negative result; do not relabel any of these post-outcome controls as a new success.

Methods, numerical provenance, and limitations

Virial and conditional Jeans inputs are the unchanged disk/bulge geometry, distance/radius/luminosity, M/L=2, Plummer and tapered-cusp profiles of v7.05-v7.06. The v7.07 first-gradient control replaced only the field law and independently evaluated angular/radial integrals. The massive-field result follows algebraic Fourier partial fractions. The planar result follows a 2-D flux integral, with one explicitly illustrative mass normalization. The 3-D scalar results follow spherical flux, exact implicit algebraic roots, and a first-order Keplerian Gauss integral using 16,385 true-anomaly points (with 257-4,097 point checks for Mercury). The constants used were $\xi_0=1.082401\times10^{-10}$ m s^{-2}, $GM_{Sun}=1.32712440018\times10^{20}$ m^3 s^{-2}, 1 AU=149597870700 m, and Julian century=36525 days. Mercury a=0.38709893 AU, e=0.205630, orbital period=87.9691 days. Earth/Mars/Saturn semimajor axes and periods were used only for the shown illustrative perturbative estimates.

Limitations: no new 3-D nonlinear external-field boundary-value solution, no full conservation simulation, no Solar-System fit, no Cassini Q_2 prediction, no galaxy-scale projection kernel selected from physics, no photon metric or lensing, and no unique separation from existing MOND actions. A positive static ellipticity eigenvalue does not prove a causal, ghost-free relativistic theory. The isolated-spherical speed scaling is a necessary asymptotic gate, not a fit to SPARC. Post-result variants were not preregistered.

References

[1] UDCT v7.06, Post-v7.05 Action and Jeans Audit (29 Sep 2026), direct predecessor. [2] UDCT v5.32, Coupled Matter-B-Gravity Feedback Audit (24 Sep 2026), synthetic 1-D on/off trajectory test. [3] UDCT v5.33, Self-Force and Feedback Conservation Audit (24 Sep 2026), numerical sensitivity of the previous toy. [4] M. Milgrom, "Quasi-linear formulation of MOND," MNRAS 403, 886 (2010), doi:10.1111/j.1365-2966.2009.16184.x. [5] The Yukawa derivation here is direct from the written action; see, for context, scalar fifth-force derivations in Eur. Phys. J. C 84 (2024), doi:10.1140/epjc/s10052-024-12723-8. [6] S. McGaugh, F. Lelli and J. Schombert, "The Radial Acceleration Relation in Rotationally Supported Galaxies," Phys. Rev. Lett. 117, 201101 (2016), doi:10.1103/PhysRevLett.117.201101; SPARC data https://astroweb.cwru.edu/SPARC/. [7] C. M. Will, "A new general relativistic contribution to Mercury's perihelion advance" (2018), arXiv:1802.05304; L. Iorio, EPM2017 formal/realistic uncertainty discussion, arXiv:1810.13415. [8] J. Bekenstein and M. Milgrom, "Does the missing mass problem signal the breakdown of Newtonian gravity?" ApJ 286, 7 (1984), doi:10.1086/162570; the standard $\mu(x)=x/\sqrt{(1+x^2)}$ is pre-existing. [9] R. S. Park, A. Hees, B. Famaey, H. Desmond and A. Durakovic, "Improved constraints on modified Newtonian gravity from Cassini radio tracking data" (2026), arXiv:2602.17884.

End of v7.07. Structural exclusions and one corrected Solar radial control; no observational confirmation or unique UDCT bidirectional projection.